

Q find x

$$6 \leq x^2 + x \leq 12$$

Soln

$$6 \leq x^2 + x \leq 12$$

and

$$6 \leq x^2 + x$$

$$-x^2 - x + 6 \leq 0$$

$$x^2 + x - 6 \geq 0$$

$$(x+3)(x-2) \geq 0$$



$$x \in (-\infty, -3] \cup [2, \infty)$$

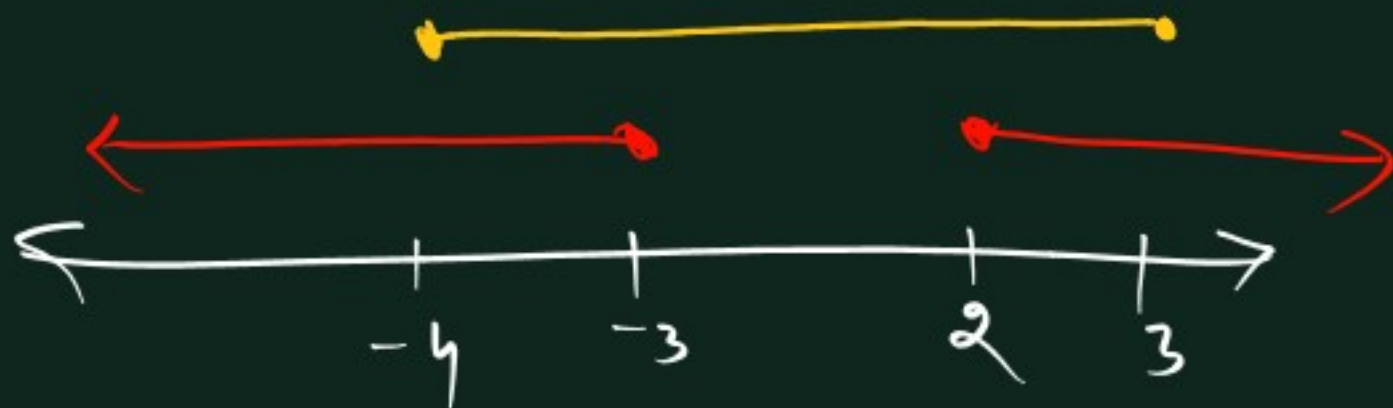
$$x^2 + x \leq 12$$

$$x^2 + x - 12 \leq 0$$

$$(x+4)(x-3) \leq 0$$



$$x \in [-4, 3]$$



$$x \in [-4, -3] \cup [2, 3]$$

Forming eq when roots are given

eg roots of the eq $ax^2+bx+c=0$ are α, β , then form the eq whose

roots are $\frac{1}{\alpha}, \frac{1}{\beta}$.

Soln new eq $\rightarrow \frac{1}{\alpha}, \frac{1}{\beta}$

$$S = \frac{1}{\alpha} + \frac{1}{\beta}$$

$$= \frac{\alpha + \beta}{\alpha\beta}$$

$$= \frac{-b/a}{c/a}$$

$$= \frac{-b}{c}$$

$$P = \frac{1}{\alpha} \times \frac{1}{\beta}$$

$$= \frac{1}{\alpha\beta}$$

$$= \frac{1}{c/a}$$

$$= \frac{a}{c}$$

$$\begin{aligned} \text{eq} \rightarrow x^2 - Sx + P &= 0 \\ x^2 + \frac{b}{c}x + \frac{a}{c} &= 0 \end{aligned}$$

$$cx^2 + bx + a = 0$$

$$ax^2 + bx + c = 0 \begin{cases} \nearrow \alpha \\ \searrow \beta \end{cases}$$

$$cx^2 + bx + a = 0 \begin{cases} \nearrow \frac{1}{\alpha} \\ \searrow \frac{1}{\beta} \end{cases}$$

$$ax^2 - bx + c = 0 \begin{cases} \nearrow -\alpha \\ \searrow -\beta \end{cases}$$

$$cx^2 - bx + a = 0 \begin{cases} \nearrow -\frac{1}{\alpha} \\ \searrow -\frac{1}{\beta} \end{cases}$$

Nature of roots ($ax^2+bx+c=0$)

- ① $a, b, c \in \mathbb{R}$, if
- $D > 0 \rightarrow$ distinct and real
 - $D = 0 \rightarrow$ equal and real
 - $D < 0 \rightarrow$ imaginary roots
 - $D \geq 0 \rightarrow$ real roots

- ② $a, b, c \in \mathbb{R}$ and $D < 0$, roots are in form of $m+in$ and $m-in$ (roots exists in pair)

- ③ $a, b, c \in \text{Rational}(\mathbb{Q})$ and $D = \text{perfect square}$.

roots are rational

$\rightarrow$ if $D \neq \text{perfect square} \Rightarrow$ roots are irrational, they will exist in pair $\Rightarrow m+\sqrt{n}, m-\sqrt{n}$

$$\sqrt{-1} = i$$
$$-1 = i^2$$

$$\sqrt{-4} = 2\sqrt{-1} = 2i$$

$$\sqrt{-5} = \sqrt{5}\sqrt{-1} = \sqrt{5}i$$

④ if $a=1$, $b, c \in \text{integer}$, and $D = \text{perfect square}$ then roots are integer.

⑤ if a, b, c are all odd integer, then roots will always be irrational given $D > 0$.

Q Find value of a if roots of the eq $x^2 - 4ax + (a+5) = 0$ are real and distinct.

Q Find the quadratic eq whose coeff are real and one root is $2i-1$. Also find the other root.

Q Find the quadratic eq whose coeff are rational and one root is $\sqrt{2}-1$. Also find the other root.

Q If the roots of the eq $x^2 + 7x + a = 0$ has integer roots then find positive values of a .

~~Q~~ for 3 quad eq $ax^2 + 2bx + c = 0$
 $bx^2 + 2cx + a = 0$
 $cx^2 + 2ax + b = 0$

- ☒ A) all of them has real roots
- ☒ B) None of them has real roots
- ☒ C) atleast one of them has real roots
- ☒ D) atmost one of them has real roots.

Soln

$$D_1 = 4b^2 - 4ac = 4(b^2 - ac)$$
$$D_2 = 4c^2 - 4ab = 4(c^2 - ab)$$
$$D_3 = 4a^2 - 4bc = 4(a^2 - bc)$$
$$D_1 + D_2 + D_3 = 4[a^2 + b^2 + c^2 - ab - bc - ca]$$
$$= 4 \times \frac{1}{2} [(a-b)^2 + (b-c)^2 + (c-a)^2]$$
$$= 2 [(a-b)^2 + (b-c)^2 + (c-a)^2]$$

≥ 0

Soln |

$$x^2 - 4ax + (a+5) = 0$$

distinct and real $\rightarrow D > 0$

$$(-4a)^2 - 4 \cdot 1 \cdot (a+5) > 0$$

$$16a^2 - 4a - 20 > 0$$

$$4a^2 - a - 5 > 0$$

$$4a^2 - 5a + 4a - 5 > 0$$

$$a(4a-5) + 1(4a-5) > 0$$

$$(4a-5)(a+1) > 0$$

$$a \in (-\infty, -1) \cup \left(\frac{5}{4}, \infty\right)$$



Soln

coeff are real, one root is $2i-1$
other root is $-2i-1$

$$S = (\cancel{2i}-1) + (\cancel{-2i}-1) \\ = -2$$

$$P = (2i-1)(-2i-1) \\ = -4i^2 - \cancel{2i} + \cancel{2i} + 1 \\ \left. \begin{array}{l} i^2 = -1 \\ \end{array} \right\} = 5$$

$$\text{eq} \rightarrow \begin{array}{l} x^2 - Sx + P = 0 \\ x^2 + 2x + 5 = 0 \end{array}$$

m2

$$x = 2i - 1$$

$$x + 1 = 2i$$

$$(x+1)^2 = (2i)^2$$

$$x^2 + 2x + 1 = 4i^2$$

$$x^2 + 2x + 5 = 0$$

Soln

coeff are rational, one root is $\sqrt{2}-1$
other root is $-\sqrt{2}-1$

$$x = \sqrt{2} - 1$$

$$x+1 = \sqrt{2}$$

$$(x+1)^2 = \sqrt{2}^2$$

$$x^2 + 2x + 1 = 2$$

$$x^2 + 2x - 1 = 0$$

Soln

$$x^2 + 7x + a = 0 \text{ integer roots}$$

$$a \text{ is integer} \mid \begin{array}{l} D = 49 - 4a \\ 49 - 4a \text{ is perfect sq} \end{array}$$

$$a = 6, D = 25 \checkmark$$

$$a = 10, D = 9 \checkmark$$

$$a = 12, D = 1 \checkmark$$

$$a = \{6, 10, 12\}$$

Common roots

① both roots common

$$ax^2 + bx + c = 0 \quad \swarrow \begin{matrix} \alpha \\ \beta \end{matrix}$$

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha \cdot \beta = -\frac{c}{a}$$

$$px^2 + qx + r = 0 \quad \swarrow \begin{matrix} \alpha \\ \beta \end{matrix}$$

$$\alpha + \beta = -\frac{q}{p}$$

$$\alpha \cdot \beta = -\frac{r}{p}$$

$$\frac{-b}{a} = -\frac{q}{p}$$

$$\frac{p}{a} = \frac{q}{b}$$

$$\frac{c}{a} = -\frac{r}{p}$$

$$\frac{p}{a} = \frac{r}{c}$$

Condition for 2 common roots

$$\boxed{\frac{p}{a} = \frac{q}{b} = \frac{r}{c}}$$

Q If the eq $(a+b)x^2 + 5x + 10 = 0$ and $10x^2 + (a-b)x + 20 = 0$ has 2 common roots, find a and b .

$$(a+b)x^2 + 5x + 10 = 0$$

$$10x^2 + (a-b)x + 20 = 0$$

$$\frac{a+b}{10} = \frac{5}{a-b} = \frac{10}{20}$$

$$\left\{ \begin{array}{l} \frac{a+b}{10} = \frac{10}{20} \\ 2a + 2b = 10 \\ a + b = 5 \end{array} \right\} \left\{ \begin{array}{l} \frac{5}{a-b} = \frac{10}{20} \\ 10 = a-b \\ a - b = 10 \end{array} \right.$$

$$\therefore a = 7.5, b = -2.5$$

1 common root

$$a_1x^2 + b_1x + c_1 = 0 \quad | \quad a_2x^2 + b_2x + c_2 = 0$$

condition $\rightarrow$
$$\begin{vmatrix} a_1 & a_2 \\ c_1 & c_2 \end{vmatrix}^2 = \begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix} \begin{vmatrix} b_1 & b_2 \\ c_1 & c_2 \end{vmatrix}$$

method 2

- $\rightarrow$ assume the common root to be α . and put it in both the eq.
- $\rightarrow$ eliminate the highest degree term (of α) and find value of α .
- $\rightarrow$ put the value of α in one of those eq.

Determinant

$$\begin{vmatrix} 2 & 3 \\ 4 & 5 \end{vmatrix} = 10 - 12 = -2$$

$$\begin{vmatrix} x & y \\ 2 & 3 \end{vmatrix} = 3x - 2y$$

Q If the eq $x^2 - 4x + k = 0$ and $x^2 - 6x + (2k-1) = 0$ has a common root, then find value of k and also find the corresponding common root.

Soln Let the common root α

$$\alpha^2 - 4\alpha + k = 0 \quad \text{--- (1)}$$

$$(-) \quad \alpha^2 - 6\alpha + (2k-1) = 0 \quad \text{--- (2)}$$

$$\hline 2\alpha + (-k+1) = 0$$

$$\alpha = \frac{k-1}{2}$$

$$\alpha = \frac{k-1}{2} \text{ in (1)}$$

$$\left(\frac{k-1}{2}\right)^2 - 4\left(\frac{k-1}{2}\right) + k = 0$$

$$\Rightarrow \frac{k^2 - 2k + 1}{4} - 2k + 2 + k = 0$$

$$\Rightarrow k^2 - 2k + 1 - 8k + 8 + 4k = 0$$

$$\Rightarrow k^2 - 6k + 9 = 0$$

$$(k-3)^2 = 0$$

$$k = 3$$

$$\therefore \alpha = \frac{3-1}{2} = 1$$

Q If the eq $x^2 - 3x + k - 1 = 0$ and $x^2 - 5x + 2k = 0$ has a common root, find k and also corresponding common root.

Soln

$$\begin{array}{l|l} x^2 - 3x + k - 1 = 0 & x = \frac{k+1}{2} \text{ in } \textcircled{1} \\ (-) \quad x^2 - 5x + 2k = 0 & \left(\frac{k+1}{2}\right)^2 - 3\left(\frac{k+1}{2}\right) + (k-1) = 0 \\ \hline 2x - k - 1 = 0 & \frac{k^2 + 2k + 1}{4} - \frac{3k + 3}{2} + k - 1 = 0 \\ x = \frac{k+1}{2} & \end{array}$$

$$k^2 + \cancel{2k} + 1 - \cancel{6k} - 6 + \cancel{4k} - 4 = 0$$

$$k^2 - 9 = 0$$

$$k = +3$$

$$x = 2$$

$$\begin{array}{l|l} \text{or } k = -3 & \\ \hline x = \frac{-3+1}{2} & \\ & = -1 \end{array}$$

Q If the eq $x^2 + 3x + 6 = 0$ and $2x^2 + ax + b = 0$ has a root in common, find value of a and b . ($a, b \in \mathbb{R}$)

Soln

$$x^2 + 3x + 6 = 0$$

$$D = 9 - 24 = -15$$

imaginary roots

$m+in, m-in$

Common root = $m+in$
 $m-in$

$$2x^2 + ax + b = 0$$

$\frac{m+in, m-in}{m-in, m+in}$

$$\frac{1}{2} = \frac{3}{a} = \frac{6}{b}$$

$$a = 6, b = 12$$

→ Coeff are real, if one the eq has imaginary roots and they have a common root then actually they have 2 common root. and condition of 2 common root to be applied.

→ if Coeff are rational, if one of the eq has irrational roots and they have a common root, they actually have 2 common root and condition of 2 common root to be applied.

Q the eq $ax^2 - 2bx + 5 = 0$ has equal roots and it has a common root with $x^2 - 2bx - 10 = 0$, then find value of b^2 .

Soln $ax^2 - 2bx + 5 = 0$ $\swarrow \searrow$

$$\alpha + \alpha = \frac{2b}{a} \quad | \quad \alpha \cdot \alpha = \frac{5}{a}$$

$$\alpha = \frac{b}{a} \quad | \quad \alpha^2 = \frac{5}{a}$$

$$\frac{b^2}{a^2} = \frac{5}{a}$$

$$b^2 = 5a$$

$$b^2 = \frac{5}{4}$$

$$x^2 - 2bx - 10 = 0 \quad \swarrow \searrow$$

$$\alpha + \beta = 2b, \quad | \quad \alpha\beta = -10$$

$$\frac{b}{a} + \beta = 2b$$

$$\frac{b\beta}{a} = -10$$

$$\beta = 2b - \frac{b}{a}$$

$$\frac{b}{a} \left(2b - \frac{b}{a} \right) = -10$$

$$\frac{2b^2}{a} - \frac{b^2}{a^2} = -10$$

$$\frac{100}{a} - \frac{5a}{a^2} = -10$$

$$\frac{+5}{a} = +20$$

$$a = \frac{5}{20}$$

$$= \frac{1}{4}$$

theory of eq (vieta's relation)

$$ax^2 + bx + c = 0 \begin{matrix} \swarrow \alpha \\ \searrow \beta \end{matrix} \quad \bigg| \quad ax^3 + bx^2 + cx + d = 0 \begin{matrix} \swarrow \alpha \\ \searrow \beta \\ \quad \gamma \end{matrix}$$

$$\alpha + \beta = \frac{-b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

$$\alpha + \beta + \gamma = \frac{-b}{a}$$

$$\alpha\beta + \beta\gamma + \gamma\alpha = \frac{c}{a}$$

$$\alpha\beta\gamma = \frac{-d}{a}$$

$$ax^4 + bx^3 + cx^2 + dx + e = 0 \begin{matrix} \swarrow \alpha \\ \searrow \beta \\ \quad \gamma \\ \quad \quad s \end{matrix}$$

$$\alpha + \beta + \gamma + s = \frac{-b}{a}$$

$$\alpha\beta + \alpha\gamma + \alpha s + \beta\gamma + \beta s + \gamma s = \frac{c}{a}$$

$$\alpha\beta\gamma + \alpha\beta s + \alpha\gamma s + \beta\gamma s = \frac{-d}{a}$$

$$\alpha\beta\gamma s = \frac{e}{a}$$

$$\text{sum of roots} = \frac{-\text{coeff of } x^{n-1}}{\text{coeff of } x^n}$$

$$\text{product of roots} = \frac{(-1)^n \text{ constant term}}{\text{coeff of } x^n}$$

degree = n

sum of product of roots taken 3 at a time.

Q roots of the eq $x^3 - x^2 + 2x + 3 = 0$ are α, β, γ . find

(i) $\alpha^2 + \beta^2 + \gamma^2 = ?$

(ii) $\alpha^3 + \beta^3 + \gamma^3 = ?$

(iii) $(\alpha + \beta)(\beta + \gamma)(\gamma + \alpha) = ?$

Soln

$$\alpha + \beta + \gamma = 1$$

$$\alpha\beta + \beta\gamma + \gamma\alpha = 2$$

$$\alpha\beta\gamma = -3$$

$$\begin{aligned} \text{(i)} \quad \alpha^2 + \beta^2 + \gamma^2 &= (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \beta\gamma + \gamma\alpha) \\ &= 1^2 - 2 \times 2 \\ &= -3 \end{aligned}$$

$$\text{(ii)} \quad \underbrace{\alpha^3 + \beta^3 + \gamma^3} - 3\underbrace{\alpha\beta\gamma} = (\alpha + \beta + \gamma)(\underbrace{\alpha^2 + \beta^2 + \gamma^2} - \underbrace{\alpha\beta - \beta\gamma - \gamma\alpha})$$

$$\Rightarrow \alpha^3 + \beta^3 + \gamma^3 + 9 = 1(-3 - 2)$$

$$\Rightarrow \alpha^3 + \beta^3 + \gamma^3 = -14$$

$$\begin{aligned}
& (\alpha + \beta)(\beta + \gamma)(\gamma + \alpha) \\
&= (1 - \gamma)(1 - \alpha)(1 - \beta) \\
&= (1 - \gamma)[1 - \beta - \alpha + \alpha\beta] \\
&= (1 - \beta - \alpha + \alpha\beta - \gamma + \beta\gamma + \gamma\alpha - \alpha\beta\gamma) \\
&= (1 - \alpha - \beta - \gamma + \alpha\beta + \beta\gamma + \gamma\alpha - \alpha\beta\gamma) \\
&= (1 - 1 + 2 + 3) \\
&= 5
\end{aligned}$$

Q If roots of the eq $x^3 - 6x^2 + 11x + p = 0$ are in AP, find $p = ?$

$$x^3 - 6x^2 + 11x + p = 0 \begin{cases} \nearrow a-d \\ \rightarrow a \\ \searrow a+d \end{cases}$$

$$\begin{aligned} \cancel{(a-d)} + (a) + \cancel{(a+d)} &= 6 \\ 3a &= 6 \\ a &= 2 \end{aligned} \quad \left| \begin{aligned} (a-d)a + a(a+d) + (a+d)(a-d) &= 11 \\ \cancel{a^2 - ad} + \cancel{a^2 + ad} + a^2 - d^2 &= 11 \\ 3a^2 - d^2 &= 11 \\ 12 - d^2 &= 11 \\ d^2 &= 1 \end{aligned} \right| \quad \begin{aligned} (a-d)a(a+d) &= -p \\ (2-d)2(2+d) &= -p \\ 2(4-d^2) &= -p \\ 2(3) &= -p \\ p &= -6 \end{aligned}$$